

Title: Resolving the Quantum Measurement Gap via the Unified Resonance-Torsion Framework

Author: Branden Lee Friend

1. Introduction: The Quantum Measurement Gap

In standard quantum mechanics, the evolution of a system is governed by the Schrödinger equation, which is linear, deterministic, and unitary:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

However, during a measurement, the system appears to abruptly "collapse" into an eigenstate of the observable, violating this smooth evolution. This discontinuity between deterministic evolution and probabilistic measurement is the **quantum gap**.

Traditionally, interpretations like Copenhagen (collapse), Many-Worlds (branching), and decoherence attempt to address this, but none resolve the core problem within a fully physical, unified field-theoretic context.

2. The Resonance-Torsion Solution: Physical Overview

This framework eliminates the need for collapse by introducing a physically grounded dynamical mechanism:

Measurement is a process of resonance lock-in

When an observer interacts with a quantum system, the scalar field $\phi(x, t)$ (which permeates spacetime and couples to matter) dynamically responds to ambient energy, spin, and information gradients. This scalar field is coupled to spacetime torsion $T_{\mu\nu\lambda}$, an antisymmetric geometric field capturing intrinsic spin density and rotational degrees of freedom in the affine connection.

The measurement event triggers a **resonance cascade** between ϕ , $T_{\mu\nu\lambda}$, and the observer information gradient $\nabla_\mu I(x)$, causing the system to converge to a stable eigenmode rather than stochastically collapsing.

3. The Mathematical Structure

3.1 Unified Action Functional

$$S = \int d^4x \sqrt{-g} \mathcal{L}_{\text{total}}$$

Where the Lagrangian includes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gravity}} + \mathcal{L}_{\phi} + \mathcal{L}_{\text{torsion}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{feedback}} + \dots$$

3.2 Key Terms

- **Gravity:**

$$\mathcal{L}_{\text{gravity}} = \frac{1}{16\pi G}(R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \gamma R\Box R)$$

- **Scalar Field (ϕ):**

$$\mathcal{L}_{\phi} = \frac{1}{2}\partial_{\mu}\phi\partial^{\mu}\phi - V(\phi), \quad V(\phi) = \frac{1}{2}m^2\phi^2 + \lambda_{\phi}\phi^4 + \xi\phi T$$

- **Torsion:**

$$\mathcal{L}_{\text{torsion}} = \eta S_{\mu\nu\lambda}S^{\lambda\mu\nu}, \quad T_{\mu\nu}{}^{\lambda} = S_{\mu\nu}{}^{\lambda}$$

- **Observer Feedback:**

$$\mathcal{L}_{\text{feedback}} = \zeta\phi T_{\lambda\mu\nu}T^{\lambda\mu\nu} + \chi g_{\mu\nu}\nabla^{\mu}I\nabla^{\nu}I + g_{\tau}J_{\mu}^5u^{\mu}$$

4. The Measurement Mechanism

4.1 Field Equation for ϕ

$$\Box\phi - m^2\phi = \lambda T + \xi T^{\mu\nu\lambda}T_{\mu\nu\lambda} + \dots$$

- The ϕ -field responds to mass-energy (T), spin (via torsion), and resonance terms.
- It behaves as a *field that converges* to eigenmodes under environmental influence.

4.2 Resonance Lock-in

Measurement corresponds to ϕ converging to a mode ϕ_n that satisfies the coupled system:

$$\Box\phi_n = \lambda T + \text{torsion feedback} \Rightarrow \phi_n \sim \text{eigenmode selected by boundary and spectral conditions}$$

This is mathematically supported by a **Fredholm determinant structure**:

$$\det(I - H_{\delta}(s)) = \zeta(s + \delta)$$

indicating that all physical fields are constrained by number-theoretic resonance spectra.

5. Resolution of the Measurement Problem

In summary: - The observer interacts with the system via information gradients $\nabla I(x)$. - This modifies spacetime via $g_{\mu\nu} \sim \nabla_\mu I \nabla_\nu I$. - The scalar ϕ and torsion $T_{\mu\nu\lambda}$ fields respond dynamically. - Rather than collapse, the system **locks into a resonant eigenstate**. - This process respects energy conservation and unitarity.

There is no stochastic projection — measurement is a **deterministic resonance convergence** governed by physical fields.

6. Implications

- This framework restores determinism and field causality to quantum mechanics.
- It integrates GR, QFT, and quantum measurement into one action.
- It proposes observable fifth-force effects, vacuum polarization, and quantized coupling corrections.

The quantum gap is closed not by interpretation, but by **explicit field dynamics**, anchored in geometry, torsion, scalar flows, and observer interaction.

Contact: brandenfriend007@outlook.com

All Rights Reserved by Owner